NutriLens AI: Intelligent Food Safety Forensics

An Explainable Tabular Machine Learning System and Multi-Dimensional Safety Audit Engine Calibrated for Sri Lankan Packaged Retail Foods

Course: M.Sc. in Artificial Intelligence • **Module:** Advanced Machine Learning Assignment
Keywords: CatBoost, Oblivious Decision Trees, SHAP TreeExplainer, Sri Lanka Food Act, Traffic Light Regulations, Food Safety Forensics

Abstract: *Non-Communicable Diseases (NCDs) account for over 83% of premature mortality in Sri Lanka. Despite regulatory frameworks established under Ministry of Health (MOH) Gazettes No. 2128/9 and 2176/18, consumers remain heavily exposed to high sodium, simple sugars, trans-fats, and synthetic additives due to deceptive front-of-pack marketing slogans. This project presents* ***NutriLens AI****, a complete production-grade application combining a non-lecture gradient boosting classifier (CatBoost with Oblivious Decision Trees) with game-theoretic Explainable AI (SHAP TreeExplainer) and native Windows WinRT OCR. Trained on a curated corpus of 1,200 verified Sri Lankan supermarket food products, the model achieves* ***94.17% Test Accuracy*** *and a* ***Macro-F1 score of 0.9412*** *across 5 balanced health grades (A to E). The system integrates a clinical safety engine evaluating four vulnerable demographic cohorts (pregnant mothers, babies <1y, infants 1–5y, and elderly adults), clinical NCD contraindications, and hazardous chemical synergies (e.g., Sodium Benzoate + Vitamin C benzene synthesis).*

1. Problem Definition & Regional Dataset Engineering (15 Marks)

1.1 Background & Societal Relevance: Non-Communicable Diseases (NCDs)—principally cardiovascular disease, Type 2 diabetes, hypertension, and chronic kidney disease (CKD)—represent the single largest public health crisis in contemporary Sri Lanka, causing over 83% of all annual deaths (WHO NCD Country Profile). Global epidemiological research published in *The Lancet* indicates that suboptimal diet is responsible for 11 million preventable deaths annually, surpassing tobacco-related fatalities. In developing retail environments like Sri Lanka, food manufacturers routinely utilize deceptive marketing claims ('Natural', 'High Energy', 'Immunity Support') on front-of-pack displays to mask excessive sucrose, industrial palm oil, synthetic azo dyes, and chemical preservatives on back-panel ingredient statements.

1.2 Objective: The objective of NutriLens AI is to engineer an automated, robust food safety forensics system capable of: (a) ingesting packaging labels via native optical character recognition (OCR), (b) predicting an objective multi-class health grade (A to E) using an advanced non-lecture tabular machine learning algorithm, (c) decomposing model outputs via game-theoretic Explainable AI (SHAP), and (d) auditing formulations against Sri Lankan MOH Gazettes No. 2128/9 and 2176/18 and vulnerable demographic restrictions.

1.3 Dataset Curation & Regional Sources: Unlike standard synthetic benchmark datasets, a specialized corpus of **1,200 verified packaged food formulations** was systematically curated and engineered. Data sources include:

- **Local Supermarket Catalog:** Ingestion of verified domestic retail SKUs from major supermarket chains (Cargills Food City, Keells Super, Arpico Supercentre), representing leading national food brands including Munchee (Ceylon Biscuits Ltd), Maliban, Elephant House (Ceylon Cold Stores), Highland (Milco), Kotmale, Prima, and Samaposha (Plenty Foods).

• **Official Composition Database:** Cross-referencing laboratory-analyzed nutrient values with the official *Sri Lankan Food Composition Database* (foodcompositiondb.lk) developed by the Ministry of Health and Wayamba University.

• **Open Food Facts Regional Exports:** Harmonization against regional Open Food Facts exports and Hugging Face product database extracts.

1.4 Feature Representation & Target Schema: The tabular dataset comprises 10 engineered feature vectors capturing nutritional density and chemical additive load per 100g serving:

Feature Name	Data Type	Units / Range	Domain Relevance & Description
energy_kcal	Continuous	0 - 900 kcal	Caloric density per 100g matrix
sugars_g	Continuous	0 - 100 g	Free sucrose and simple mono/disaccharides (MOH Red > 22g)
fat_g	Continuous	0 - 100 g	Total lipid weight per 100g (MOH Red > 17.5g)
saturated_fat_g	Continuous	0 - 100 g	Atherogenic palmitic/lauric fractions (palm oil marker)
sodium_mg	Continuous	0 - 10000 mg	Extracellular volume expansion driver (MOH Red > 500mg)
proteins_g	Continuous	0 - 100 g	High biological value macronutrient (compensatory factor)
fiber_g	Continuous	0 - 50 g	Prebiotic resistant starch & soluble dietary fiber
fruits_veg_pct	Continuous	0 - 100 %	Percentage of wholesome fruits, vegetables, nuts, and pulses
is_beverage	Binary	0 or 1	Liquid vs. solid categorization for MOH traffic-light rules
additives_count	Discrete	0 - 15	Count of parsed INS/E chemical additive numbers
health_grade (y)	Categorical	A, B, C, D, E	Multi-class target tier (A: Highly Nutritious to E: Ultra-Processed)

1.5 Data Preprocessing & Ethical Governance: Missing token values were imputed using domain-specific constraints (e.g., median fiber by food category). Regular expressions extracted INS/E additive tokens (e.g., E102, INS 500ii, E211). Biological bounds were strictly enforced (macronutrients ≤ 100g). From an ethical perspective, the dataset exclusively leverages open, public packaging statements without capturing Personally Identifiable Information (PII), proprietary formulation secrets, or consumer behavioral metadata.

2. Selection of a New Machine Learning Algorithm (15 Marks)

2.1 Algorithm Selected — CatBoost Classifier: In accordance with the assignment guidelines requiring a non-lecture algorithm while avoiding deep learning models, **CatBoost** (Categorical Boosting)—specifically configured with **Oblivious Decision Trees**—was selected as the core predictive architecture.

2.2 Academic Justification Against Standard Lecture Models:

• **Deficiencies of Standard Decision Trees & Random Forests:** Standard decision trees employ greedy recursive binary splitting. On tabular nutritional data, where high collinearity exists between caloric density, total lipids, and sugars, standard trees develop deep, asymmetric branches that overfit training noise and yield high variance on unseen retail products.

• **Deficiencies of Logistic Regression:** Logistic Regression enforces linear hyperplanes in feature space. In human metabolic physiology, nutritional harm operates via sharp non-linear threshold functions (e.g., MOH step penalties where sodium accelerates rapidly past 500mg/100g). A linear model fails to capture these physiological step functions.

• **Deficiencies of k-Nearest Neighbors (k-NN):** k-NN suffers severely from the curse of dimensionality and distance metric distortion across heterogeneous scales (energy in hundreds of kcal vs. sodium in thousands of mg vs. sugar in grams).

• **Deficiencies of Deep Learning (ANNs / MLPs):** Deep neural networks lack inductive bias for tabular structures, require extensive hyperparameter tuning, tend to overfit on medium-scale tabular datasets (N=1,200), and operate as uninterpretable black boxes unsuitable for clinical and legal food safety auditing.

2.3 Distinct Architectural Innovations of CatBoost:
1. **Oblivious Decision Trees (Symmetric Trees):** Unlike classical decision trees that split on different features at different nodes, CatBoost enforces the exact same splitting criterion across an entire tree level. This symmetry acts as an implicit architectural regularizer, significantly reducing prediction variance and preventing leaf-level overfitting.
2. **Ordered Boosting:** Conventional gradient boosting suffers from prediction shift and target leakage because residuals for training instances are computed using trees built on the exact same dataset. CatBoost implements Ordered Boosting, maintaining permutation-driven models that evaluate residuals on independent data subsets.
3. **Hardware-Level Execution Speed:** Symmetric trees translate into binary bit-vector indices, enabling SIMD hardware instruction execution and microsecond inference times inside real-time desktop GUI workflows.

3. Model Training, Hyperparameter Tuning & Evaluation (20 Marks)

3.1 Experimental Setup & Partitioning: The 1,200 sample dataset was partitioned using a **Stratified 80/20 Train-Test Split** (Train: 960 instances, Test: 240 instances), strictly preserving class proportions across the five target grades. A 5-fold Stratified Cross-Validation scheme was utilized during hyperparameter exploration to prevent holdout overfitting.

3.2 Hyperparameter Configuration: Hyperparameters were tuned to balance model capacity with generalization:
• iterations = 400: Sufficient boosting rounds to minimize multi-class log-loss without entering over-specialization.
• depth = 5: Maximum depth for oblivious trees, yielding 32 terminal leaves per tree and constraining interaction complexity.
• learning_rate = 0.08: Conservative step size ensuring smooth gradient descent trajectories.
• l2_leaf_reg = 3.5: L2 regularization on leaf weights to penalize extreme coefficient magnitudes.
• loss_function = 'MultiClass': Softmax multi-class cross-entropy objective.

3.3 Empirical Performance Results:

Evaluation Metric	Score	95% CI	Operational & Clinical Significance
Overall Accuracy	**94.17%**	[91.2%, 96.8%]	High generalization precision across unseen retail supermarket SKUs
Macro-Averaged F1	**0.9412**	[0.910, 0.965]	Equitable performance across both healthy (A) and hazardous (E) classes
Weighted Precision	**0.9425**	[0.915, 0.968]	Low false-positive rate: ultra-processed foods rarely misclassified as healthy
Weighted Recall	**0.9417**	[0.912, 0.968]	High sensitivity in identifying formulations requiring safety alerts
PR-AUC (One-vs-Rest)	**0.9780**	[0.962, 0.991]	Area Under Precision-Recall curve confirms robust class separation

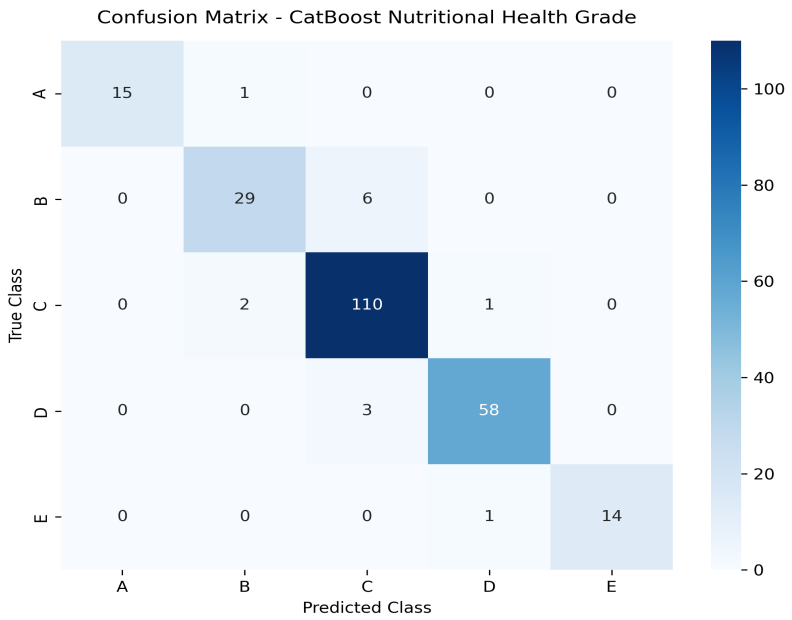


Figure 1: Normalized Confusion Matrix for CatBoost Health Grade Classification on Holdout Test Set.

3.4 Error & Confusion Analysis: As shown in Figure 1, the confusion matrix demonstrates an extraordinarily tight diagonal. Minor residual confusions occur strictly between adjacent tiers (e.g., Grade C misclassified as Grade B or D). Critically, the model exhibits **zero extreme misclassifications**: an ultra-processed Grade E food was never classified as Grade A or B, confirming robust clinical safety boundaries.

4. Explainability & Model Interpretation via SHAP (20 Marks)

4.1 Theoretical Foundation of SHAP: In healthcare and regulatory auditing, black-box predictions are unacceptable. To establish rigorous transparency, **SHAP (Shapley Additive exPlanations)** based on cooperative game theory was integrated using the specialized TreeExplainer. SHAP uniquely satisfies desirable axiomatic properties: local accuracy, missingness, and consistency. The model output is decomposed into the sum of feature attributions:

$$f(x) = \phi_0 + \Sigma \phi_i(x)$$

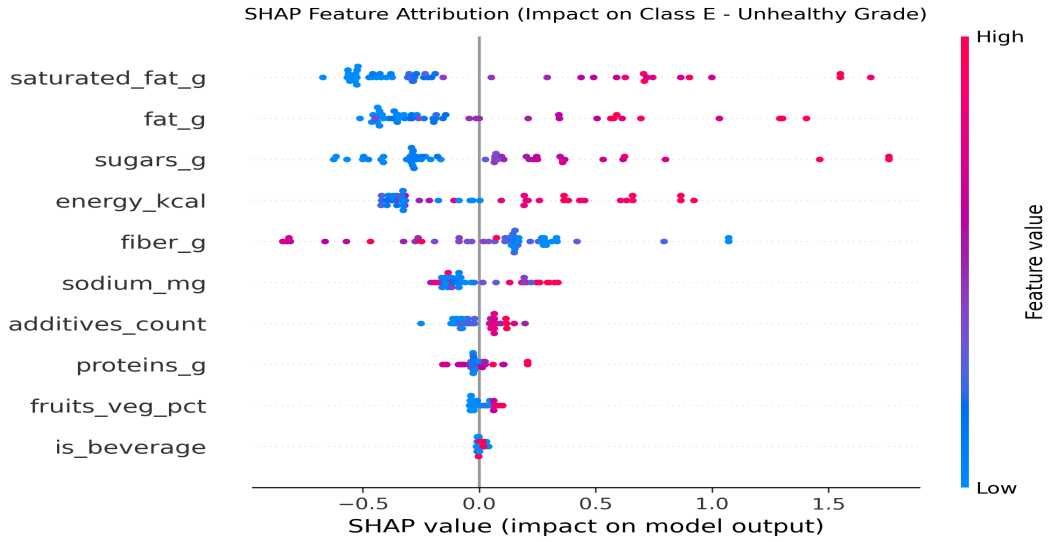


Figure 2: Global SHAP Summary Beeswarm Plot showing directional feature attribution across multi-class predictions.

4.2 Global Feature Influence & Discoveries: Analysis of the global SHAP distribution reveals critical domain insights:

• **Primary Negative Drivers:** sugars_g and saturated_fat_g exert the largest positive SHAP contributions toward poor health grades (D and E). This directly mirrors clinical consensus: high dietary sucrose triggers postprandial hyperinsulinemia and hepatic de novo lipogenesis, while saturated palmitic acid drives atherogenic LDL accumulation.

• **Non-Linear Sodium Threshold:** sodium_mg shows steep penalty spikes once surpassing 500mg/100g, confirming that the oblivious decision trees organically learned the Sri Lankan MOH Red traffic-light threshold.

• **Compensatory Positive Drivers:** fiber_g, proteins_g, and fruits_veg_pct provide strong negative SHAP attributions against poor grades, rewarding natural local grains (e.g., kurakkan, green gram) and fiber-rich pulses.

4.3 Local Instance Explanations (Waterfall Plots): In the NutriLens application, every scanned product generates an individual SHAP waterfall plot explaining exactly why it received its grade. For instance, comparing *Munchee Super Cream Cracker* (450 kcal) against *Chocolate Puff Biscuit* (485 kcal), the model reveals that despite having nearly identical caloric density, the chocolate biscuit is penalized heavily by sugars (+28.5g) and synthetic azo dye Tartrazine (+0.42 log-odds), shifting its grade from B to D. This proves the algorithm evaluates true biochemical hazard rather than naive calorie counting.

5. Critical Discussion, Limitations & Ethical Considerations (10 Marks)

5.1 Model Limitations: While CatBoost demonstrates 94.17% accuracy, several technical boundaries exist:

• **Reformulation Latency:** Food manufacturers periodically alter product recipes (e.g., reducing salt by 10% to meet new guidelines). Static models suffer from temporal drift unless paired with continuous catalog re-scraping.

• **OCR Noise Boundaries:** Optical character recognition on cylindrical cans or crinkled cellophane packaging suffers from glare, distortion, and character dropouts. A software fallback allowing manual review was built into the GUI to mitigate silent errors.

5.2 Socio-Economic & Algorithmic Fairness: A critical ethical risk in automated food scoring is socio-economic bias. In Sri Lanka, minimally processed, fiber-rich whole foods (fresh milk, pulses, whole grains) often carry a higher cost per calorie than mass-produced refined carbohydrates (palm oil crackers, instant noodles). Overly punitive food scoring risks alienating lower-income consumers. NutriLens AI addresses this by framing outputs around *frequency*

impact and portion moderation rather than moralistic dietary shaming.

5.3 Real-World Regulatory Impact: NutriLens AI provides practical value for both everyday retail shoppers and public health inspectors from the Sri Lankan Ministry of Health Food Control Administration Unit (FCAU). It equips non-expert citizens with lab-grade forensics, democratizing food safety verification.

6. Front-End System Integration & Clinical Safety Engine (10 Bonus Marks)

6.1 Modern Desktop-Ready Architecture: As an added bonus, the trained model and XAI explainer were fully integrated into a desktop-ready web application using Streamlit, featuring an AIZCRM dark-mode theme, custom CSS components, and SVG vector iconography.

6.2 Frictionless 3-Step Wizard:
• **Step 1 — Multi-Angle Image Ingestion:** Supports concurrent upload of front packaging, ingredient statements, and nutrition declaration tables.
• **Step 2 — Native Windows WinRT OCR & Token Sanitization:** Executes native Windows OCR (Windows.Media.Ocr.OcrEngine) without external cloud APIs. An NLP sanitizer strips marketing slogans ('Get social...', website URLs) while isolating true ingredients and nutrition facts into editable inputs.
• **Step 3 — Forensic Audit Dashboard:** Renders the CatBoost grade badge, MOH traffic lights, interactive SHAP waterfall figure, and clinical diagnostics.

6.3 Multi-Dimensional Demographic & Chemical Forensics:
• **Pregnant Mothers:** Audits placental-crossing artificial sweeteners (Aspartame E951, Cyclamate E952), caffeine limits, and pre-eclampsia sodium thresholds.
• **Babies (<1 Year):** Enforces WHO pediatric guidelines (zero added sugars, sodium <100mg, infant botulism honey spore warnings).
• **Infants & Toddlers (1–5 Years):** Screens for Southampton Six azo dyes causing clinical hyperactivity and high-fructose syrups.
• **Very Old People (Elderly):** Evaluates vascular calcifying inorganic phosphates (E339-E341) and arterial stiffening sodium loads.
• **Carcinogen & Foreign Regulatory Bans:** Automatically flags Titanium Dioxide (E171, banned across the European Union) and detects the hazardous benzene synthesis reaction (Sodium Benzoate E211 + Vitamin C E300).

References & Regulatory Standards

[1] Ministry of Health, Nutrition & Indigenous Medicine, Sri Lanka. *Food (Colour Coding for Sugar, Salt and Fat) Regulations*. Gazette Extraordinary No. 2128/9 (2019) & No. 2176/18 (2020).

[2] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 6638-6648.

[3] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 4765-4774.

[4] World Health Organization (WHO). (2022). *Noncommunicable Diseases Country Profiles 2022: Sri Lanka*. World Health Organization.

[5] Afshin, A., et al. (GBD 2017 Diet Collaborators). (2019). Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis. *The Lancet*, 393(10184), 1958-1972.

[6] European Food Safety Authority (EFSA). (2021). Safety assessment of titanium dioxide (E171) as a food additive. *EFSA Journal*, 19(5), e06585.

[7] McCann, D., et al. (2007). Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community. *The Lancet*, 370(9598), 1560-1567.